SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0510**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Analytical Problem Solving

Code of Conduct:

I P. Jyothika/ 192311156 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:P.Jyothika

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	

Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation s=255-r to the pixel values: r= [50,100,150,200] Compute the transformed values.

Ans:

(a) Purpose of Grey Level Transformation

Grey level transformation is used to improve the appearance of an image by changing the pixel intensity values. It helps in enhancing contrast, highlighting important features, and making image details more visible.

(b) Negative Transformation

Formula: $s = 255 - r$

r	s = 255 − r
50	205
100	155
150	105
200	55

Answer: 205, 155, 105, 55

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given s=clog(1+r), where c=10, compute s for: r= [0,10,50,100]

Ans:

(a) How it enhances low-intensity pixels

Log transformation increases the brightness of dark (low-intensity) pixels more than bright pixels. This reveals hidden details in darker regions while compressing higher intensity values.

(b) Given

$$s = 10\log(1 + r)$$

r	$s = 10\log(1 + r)$
0	0.00
10	10.41
50	17.08
100	20.04

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensi Frequen t c y y 0 2

1 2

2 4

3 8

Compute the cumulative distribution and equalized intensity values.

Ans:

(a)Role of Histogram Equalization

Histogram equalization improves image contrast by redistributing pixel intensities over the available range, making details in dark and bright regions more visible.

(b) Given Histogram

Intensity	Frequency
0	**2**

1	**2**
2	**4**
3	**8**

Total pixels $N = 2 + 2 + 4 + 8 = 16$

For a **3-bit image**, $L = 8$

Equalized intensity $s = \frac{CDF}{16} \times (8 - 1)$

Intensity	**Frequency**	**CDF**
0	2	2
1	2	4
2	4	8
3	8	16

Intensity	**Equalized Value**
0	1
1	2
2	4
3	7

- **CDF:** 2, 4, 8, 16
- **Equalized Intensities: 1, 2, 4, 7**

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region: 10 20 30 20 30 40 30 40 50 Compute the new center pixel value.

(a) Purpose

Smoothing removes noise and small variations from an image, producing a cleaner and less noisy image.

(b) Apply 3×3 Averaging Filter

Region

$$\begin{matrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{matrix}$$

Sum: $10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$

Average: $270/9 = 30$

New center pixel = 30

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask: [0 -1 0}, {-1 4 -1], { 0 -1 0] to the matrix:

10 10 10, 10 20 10, 10 10 10 Compute the output at the center.

(a) Purpose

Sharpening enhances image edges and fine details, making objects appear clearer.

(b) Laplacian Mask

$$\begin{matrix} 0 & -1 & 0 \\ [-1 & 4 & -1] \\ 0 & -1 & 0 \end{matrix}$$

Matrix

$$\begin{matrix} 10 & 10 & 10 \\ 10 & 20 & 10 \end{matrix}$$

$$10 \quad 10 \quad 10$$

Center calculation

$$(4 \times 20) - (10 + 10 + 10 + 10)$$
$$80 - 40 = 40$$

Output at the center = 40